Matching Methods

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2026-01-31

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```
bookdown::render_book()
```

Overview

Topics:

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2. Inverse Probability Weights
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Chapter 1

Propensity Score Matching

We will go through the process of manually estimating propensity scores before we use the *teffects* commands. After estimating the propensity scores, we will estimate the ATE and ATT

We will use the data from Dehejia and Wahba (2002) for treatment from Department of Labor Employment and Training Adminstration (ETA) job training. Instead of using the control group from the orginal RCT, we will use data from the Current Population Survey (CPS) for a comparison group.

1.1 Generate propensity score

First, we will import and drop the experimental control group, so we can append our comparison group from the CPS.

```
use https://github.com/scunning1975/mixtape/raw/master/nsw_mixtape.dta, clear
drop if treat==0
```

Now merge in the CPS controls from footnote 2 of Table 2 (Dehejia and Wahba 2002)

```
append using https://github.com/scunning1975/mixtape/raw/master/cps_mixtape.dta
```

Now we will generate additional variables of interest: age, education, demographics, and unemployment status

```
gen agesq=age*age
gen agecube=age*age*age
gen edusq=educ*edu
gen u74 = 0 if re74!=.
replace u74 = 1 if re74==0
gen u75 = 0 if re75!=.
replace u75 = 1 if re75==0
gen interaction1 = educ*re74
gen re74sq=re74^2
gen re75sq=re75^2
gen interaction2 = u74*hisp
```

Next, we will use logistic regression to estimate β and estimate the p-scores.

```
logit treat age agesq agecube educ edusq marr nodegree black hisp re74 re75 u74 u75 in
predict pscore
```

```
Iteration 0:   log likelihood = -1011.0713
Iteration 1:   log likelihood = -612.55814
Iteration 2:   log likelihood = -429.05652
Iteration 3:   log likelihood = -406.25926
Iteration 4:   log likelihood = -404.18813
Iteration 5:   log likelihood = -404.15992
Iteration 6:   log likelihood = -404.15991

Logistic regression                             Number of obs     =     16,177
                                                LR chi2(14)       =    1213.82
                                                Prob > chi2       =     0.0000
Log likelihood = -404.15991                     Pseudo R2         =     0.6003

------------------------------------------------------------------------------
       treat |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         age |   2.425229   .3500651     6.93   0.000     1.739114    3.111344
       agesq |  -.0672395   .0111308    -6.04   0.000    -.0890555   -.0454234
     agecube |   .0005685   .0001113     5.11   0.000     .0003505    .0007866
        educ |   .9247846   .2500694     3.70   0.000     .4346576    1.414912
       edusq |  -.0572021   .0136202    -4.20   0.000    -.0838972    -.030507
        marr |  -1.556471   .2517687    -6.18   0.000    -2.049929   -1.063013
    nodegree |   .9270591   .3254621     2.85   0.004     .2891651    1.564953
       black |   3.850668   .2662868    14.46   0.000     3.328755     4.37258
        hisp |   1.673885   .4099129     4.08   0.000     .8704704      2.4773
        re74 |  -.0002203   .0001086    -2.03   0.043    -.0004332   -7.40e-06
        re75 |  -.0001969   .0000378    -5.21   0.000     -.000271   -.0001228
         u74 |   1.749522   .2897311     6.04   0.000      1.18166    2.317385
```

```
         u75 |     .00944    .257531     0.04   0.971    -.4953114    .5141915
interaction1 |   .0000222   9.08e-06     2.45   0.014     4.43e-06      .00004
       _cons |  -35.22098   3.797922    -9.27   0.000    -42.66477   -27.77719
------------------------------------------------------------------------------
Note: 3 failures and 0 successes completely determined.

(option pr assumed; Pr(treat))
```

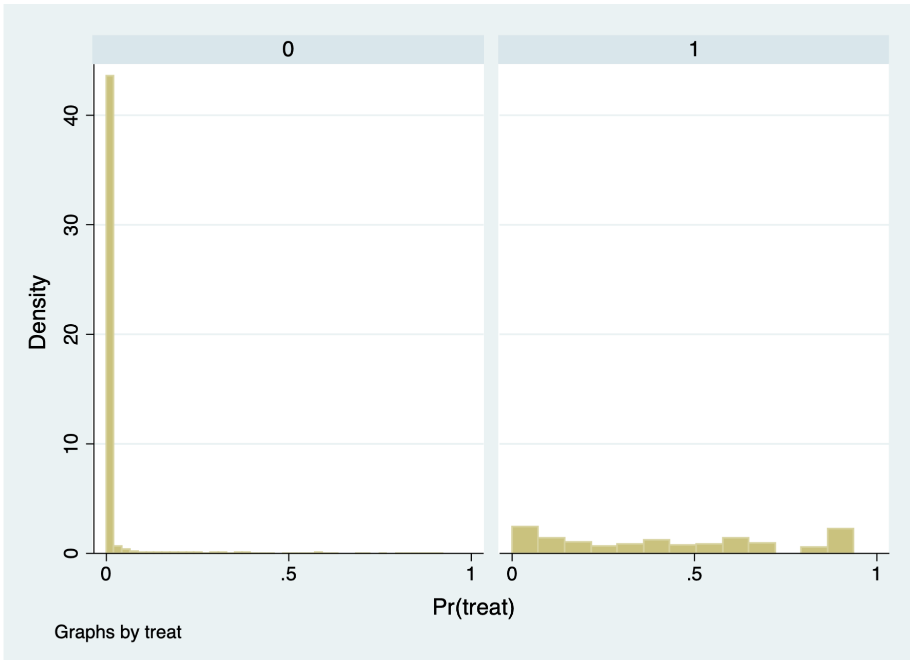
Now, we will compare the propensity score between treatment and comparison groups with summary statistics and histograms.

```
sum pscore if treat==1, detail
sum pscore if treat==0, detail
```

```
                          Pr(treat)
-------------------------------------------------------------
      Percentiles      Smallest
 1%     .0011757       .0010614
 5%     .0072641       .0011757
10%     .0260147       .0018463       Obs                 185
25%     .1322174       .0020981       Sum of Wgt.         185

50%     .4001992                      Mean           .4253567
                        Largest       Std. Dev.      .3076908
75%     .6706164       .9356451
90%     .8866026         .93718       Variance       .0946736
95%     .9021386       .9374608       Skewness       .1916478
99%     .9374608       .9384554       Kurtosis       1.732134

                          Pr(treat)
-------------------------------------------------------------
      Percentiles      Smallest
 1%     5.90e-07       1.18e-09
 5%     1.72e-06       4.07e-09
10%     3.58e-06       4.24e-09       Obs              15,992
25%     .0000193       1.55e-08       Sum of Wgt.      15,992

50%     .0001187                      Mean           .0066476
                        Largest       Std. Dev.      .0416672
75%     .0009635       .8786677
90%     .0066319       .8893389       Variance       .0017362
95%     .0163109       .9099022       Skewness        12.2493
99%     .1551548       .9239787       Kurtosis       187.8692
```

The comparison group has a mean propensity score of 0.01. The treatment group has a mean propensity score is 0.42

```
histogram pscore, by(treat) binrescale
```



It seems that there is very little overlap for the common support assumption. It would be problematic to estimate the ATE and ATT without trimming to get common support.

1.2 Estimate the ATE and ATT without trimming

We will estimate the ATE and ATT with the *teffects psmatch* command. We will use the 5 nearest neighbors with the option *nn(5)*. Note that Stata generates new variables defined by you. Here we are defining *pstub_cps*.

Estimate ATT

```
teffects psmatch (re78) (treat age agesq agecube educ edusq marr nodegree black ///
 hisp re74 re75 u74 u75 interaction1, logit), atet gen(pstub_cps) nn(5)
drop pstub_cps*
```

```
Treatment-effects estimation                   Number of obs      =     16,177
Estimator      : propensity-score matching     Matches: requested =          5
Outcome model  : matching                                     min =          5
```

```
Treatment model: logit                                         max =         26
------------------------------------------------------------------------------
             |               AI Robust
        re78 |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ATET         |
       treat |
   (1 vs 0)  |   1725.082   700.0586     2.46   0.014     352.9921    3097.172
------------------------------------------------------------------------------
```

$\widehat{ATT} = \$1,725.082$

We have a few more options for checking the balance after the PSM with nearest neighbor of 5. We can use the *teblance* commands to check our balance. *tebalance density* will give us a distribution of propensity scores by our variable of choice between treatment and comparison. We can also use *tebalance box* that will give is a box-and-whisker chart for the propensity scores by our variable of choice between treatment and comparison.

```
tebalance summarize, baseline
```

```
  Covariate balance summary
                                                 Raw      Matched
                                   -----------------------------------------
                                   Number of obs =       16,177          370
                                   Treated obs   =          185          185
                                   Control obs   =       15,992          185
                                   -----------------------------------------

  -----------------------------------------------------------------
                  |        Means               Variances
                  |    Control     Treated       Control    Treated
  ----------------+------------------------------------------------
              age |   33.22524    25.81622      121.9968    51.1943
            agesq |   1225.906    717.3946      615814.1     185978
          agecube |   49305.85    21554.66      2.04e+09   4.40e+08
             educ |   12.02751    10.34595      8.241754   4.042714
            edusq |   152.9023    111.0595      4511.316   1544.795
             marr |   .7117309    .1891892      .2051829   .1542303
         nodegree |   .2958354    .7081081      .2083299   .2078143
            black |   .0735368    .8432432      .0681334   .1329025
             hisp |    .072036    .0594595       .066851    .056228
             re74 |    14016.8    2095.574      9.16e+07   2.39e+07
             re75 |    13650.8    1532.055      8.59e+07   1.04e+07
              u74 |   .1196223    .7081081      .1053194   .2078143
              u75 |   .1093047          .6      .0973632   .2413043
```

```
    interaction1 |   171147.6    22898.73      1.67e+10   3.29e+09
  -----------------------------------------------------------------
```

After running *tebalance summarize*, we can see without trimming that our covariates are not really balanced between treatment and comparison groups.

```
tebalance density age
```

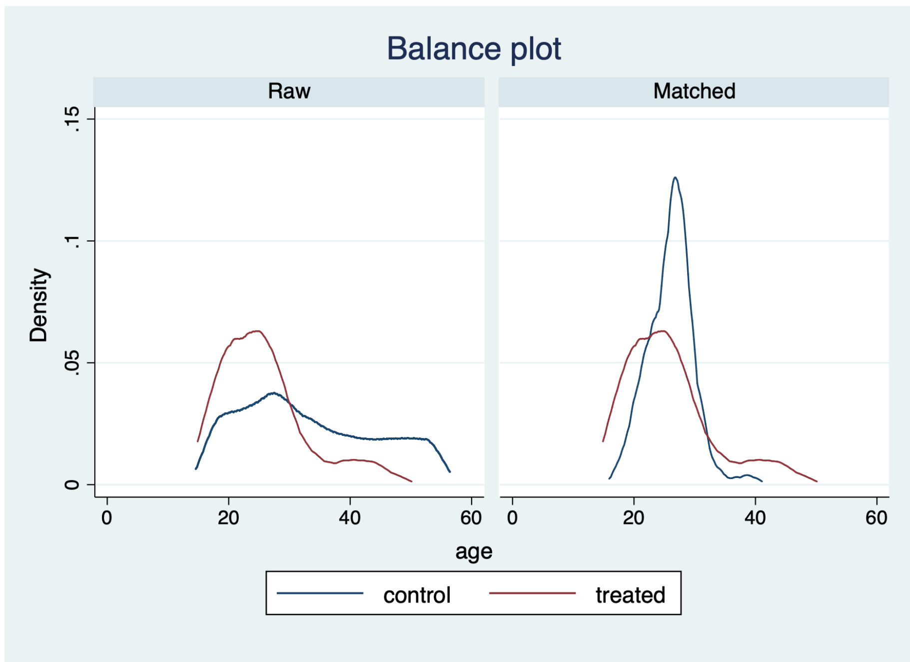


Figure 1.1: Density Chart for Age

```
tebalance box age
drop pstub_cps*
```

Next, we will estimate the *ATE* with *teffects psmatch*.

```
teffects psmatch (re78) (treat age agesq agecube educ edusq marr nodegree black ///
 hisp re74 re75 u74 u75 interaction1, logit), ate gen(pstub_cps) nn(5)

drop pstub_cps*
```

We find that we have an error estimating the *ATE*, since we lack common support.

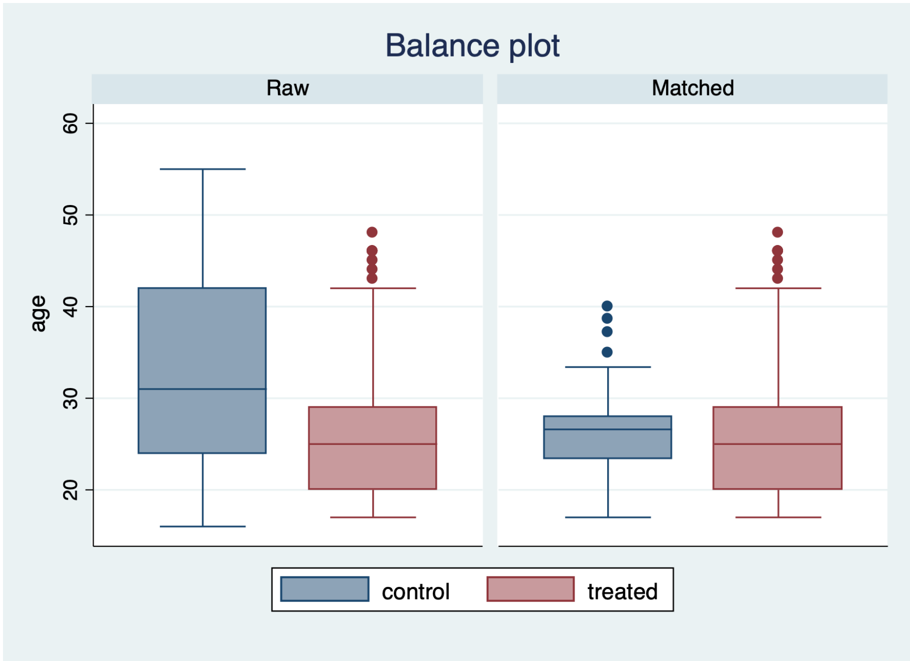


Figure 1.2: Box Chart for Age

1.3 Trim the data

We will use the trim the p-scores. All values below 10% and above 90% will be dropped.

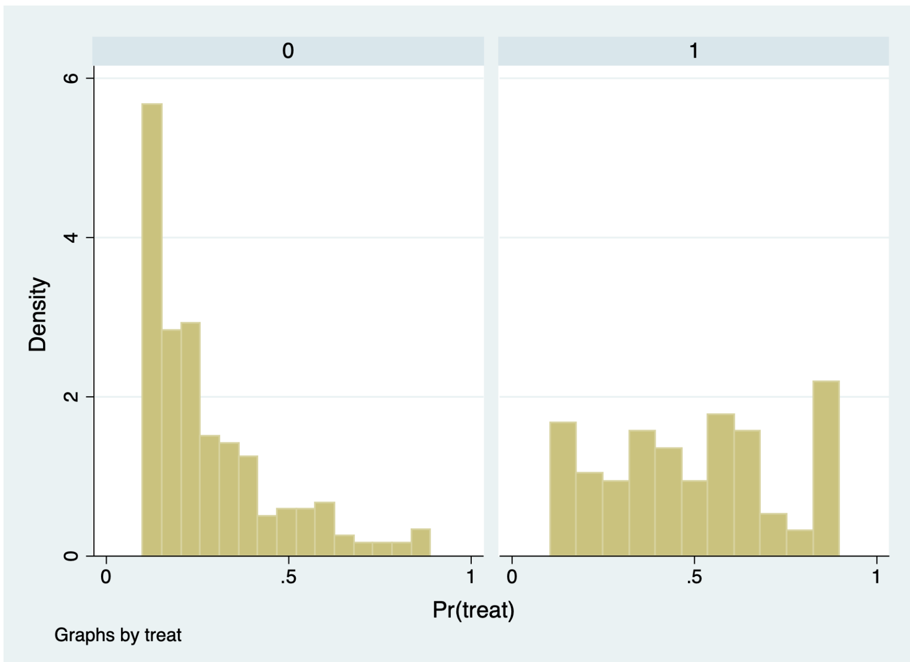
```
drop if pscore <= 0.1
drop if pscore >= 0.9
```

We can see that the comparison group has a lot of observations with p-scores less than or equal to 0.1 (15,802), but we only drop 14 observations with p-scores greater than or equal 0.9.

```
(15,802 observations deleted)

(14 observations deleted)
```

1.4 Check for balance again

```
histogram pscore, by(treat) binrescale
graph export "pscore2.png", replace
```



We can now see we have better overlap between treatment and comparison after trimming the extreme values of the p-scores.

1.5 Reestimate the ATT and ATE

```
teffects psmatch (re78) (treat age agesq agecube educ edusq marr nodegree black ///
 hisp re74 re75 u74 u75 interaction1, logit), ate gen(pstub_cps) nn(5)

drop pstub_cps*
```

```
Treatment-effects estimation                   Number of obs      =        361
Estimator      : propensity-score matching     Matches: requested =          5
Outcome model  : matching                                     min =          5
Treatment model: logit                                        max =          8
------------------------------------------------------------------------------
```

```
             |              AI Robust
        re78 |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ATE          |
       treat |
   (1 vs 0)  |   1758.837   724.5845     2.43   0.015     338.6778    3178.997
------------------------------------------------------------------------------
```

$\widehat{ATE} = \$1758.84$

```
teffects psmatch (re78) (treat age agesq agecube educ edusq marr nodegree black ///
 hisp re74 re75 u74 u75 interaction1, logit), atet gen(pstub_cps) nn(5)
```

```
Treatment-effects estimation                   Number of obs      =        361
Estimator      : propensity-score matching     Matches: requested =          5
Outcome model  : matching                                     min =          5
Treatment model: logit                                        max =          8
------------------------------------------------------------------------------
             |              AI Robust
        re78 |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ATET         |
       treat |
   (1 vs 0)  |   2519.446   866.7552     2.91   0.004     820.6375    4218.255
------------------------------------------------------------------------------
```

$\widehat{ATE} = \$2519.446$

1.6 Recheck for balance

```
tebalance summarize, baseline
```

```
  Covariate balance summary
                                                     Raw      Matched
                          -----------------------------------------
                          Number of obs =            361          266
                          Treated obs   =            133          133
                          Control obs   =            228          133
                          -----------------------------------------

  -----------------------------------------------------------------
```

```
                 |          Means                  Variances
                 |    Control     Treated       Control     Treated
-----------------+--------------------------------------------------
             age |   25.93421     25.2406      64.98684    45.00228
           agesq |   737.2851    681.7519        237422    158499.2
         agecube |   22966.88    19825.29      6.03e+08    3.66e+08
            educ |   10.46491     10.2406      4.584667    3.850763
           edusq |   114.0789    108.6917      1830.681    1331.306
            marr |   .2192982    .1428571      .1719607    .1233766
        nodegree |   .6315789    .7518797       .233712    .1879699
           black |   .9342105    .9398496       .061732    .0569606
            hisp |   .0438596     .037594      .0421207    .0364548
            re74 |   2552.555    1560.604      1.93e+07    1.56e+07
            re75 |   1570.309    1002.193       7324536     3233917
             u74 |   .5570175    .7669173       .247836    .1801094
             u75 |   .4868421    .6390977      .2509274    .2323992
    interaction1 |   27361.94    16425.15      2.44e+09    1.81e+09
--------------------------------------------------------------------
```

Our balance is greatly improved after trimming the tail ends of the propensity scores.

Let's check the variable age with *tebalance density* and *tebalance box*.

```
tebalance density age
```

```
tebalance box age
```

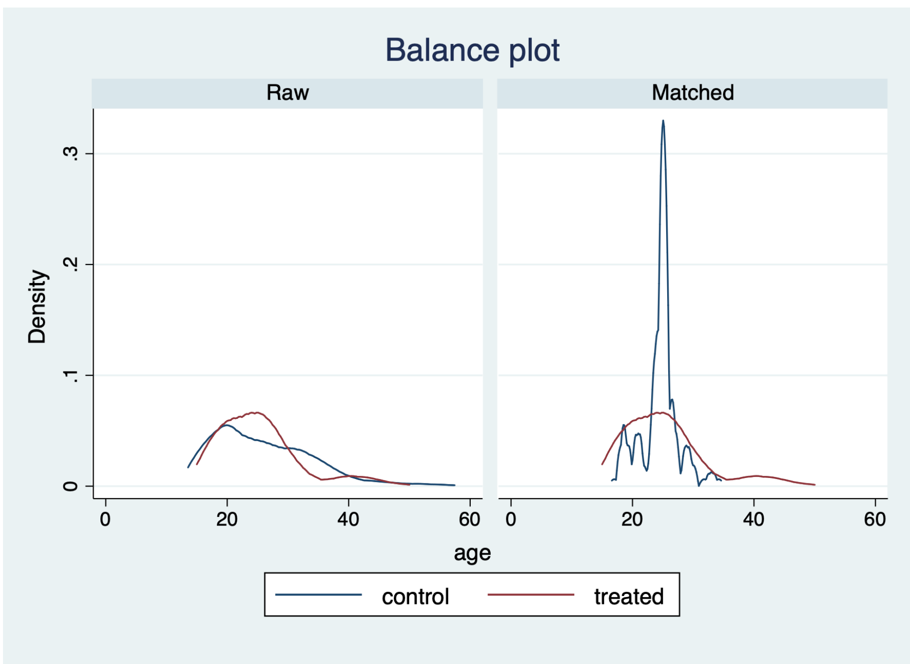


Figure 1.3: Density Chart of Age After Trimming

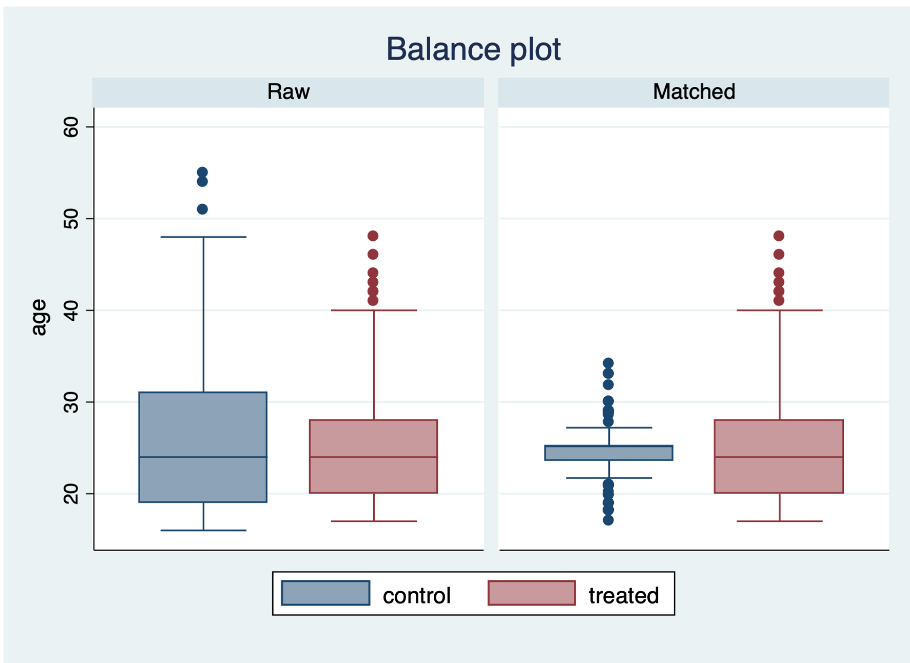


Figure 1.4: Box Chart of Age After Trimming

Chapter 2

Inverse Probability Weights

We will reload our treatment group and CPS comparison group, and use the *teffects ipw* command to estimate the ATE.

We will repull our data, estimate our p-score, and check our balance

```
*Get Data
use https://github.com/scunning1975/mixtape/raw/master/nsw_mixtape.dta, clear
drop if treat==0
append using https://github.com/scunning1975/mixtape/raw/master/cps_mixtape.dta

*Calculate Variables
gen agesq=age*age
gen agecube=age*age*age
gen edusq=educ*edu
gen u74 = 0 if re74!=.
replace u74 = 1 if re74==0
gen u75 = 0 if re75!=.
replace u75 = 1 if re75==0
gen interaction1 = educ*re74
gen re74sq=re74^2
gen re75sq=re75^2
gen interaction2 = u74*hisp

*Estimate the coefficients
logit treat age agesq agecube educ edusq marr nodegree black hisp re74 re75 u74 u75 interaction1
predict pscore

* Checking mean propensity scores for treatment and control groups
sum pscore if treat==1, detail
sum pscore if treat==0, detail
```

```
* Now look at the propensity score distribution for treatment and control groups
histogram pscore, by(treat) binrescale
```

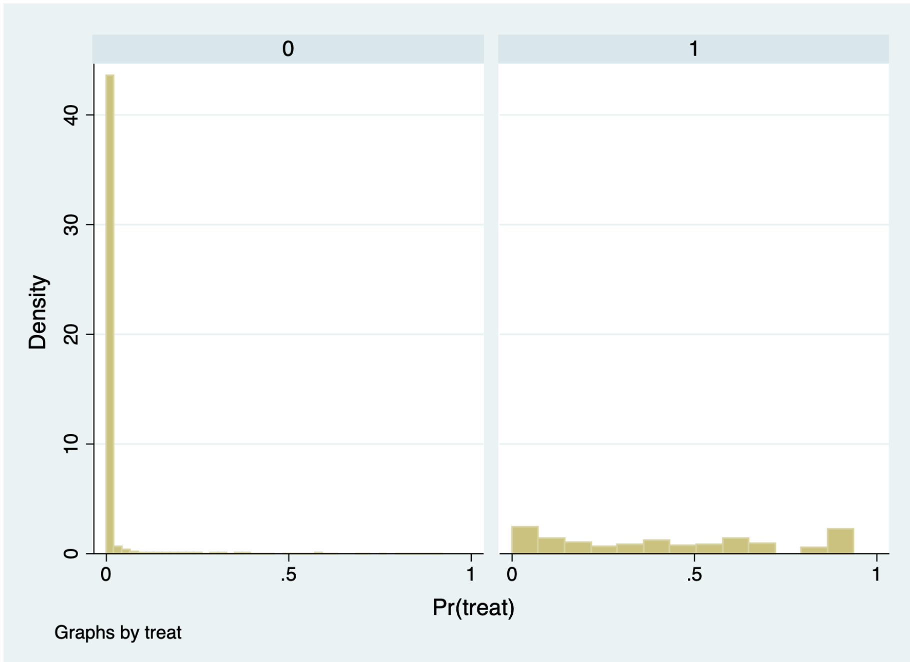


Figure 2.1: Histogram for IPW

2.1 Manual Calculate the ATE

We will manually calculate inverse probability weighting (IPW) with Propensity Scores. Please note that IPW is not matching, even though we use propensity scores.

IPW with Non-normalized Weights:

$$\widehat{ATE} = \frac{1}{n} * \sum_{n=1}^{N} \left(\frac{Y_i * D_i}{\hat{p}(x)_i} \right) - \frac{1}{n} * \sum_{n=1}^{N} \left(\frac{Y_i * (1 - D_i)}{1 - \hat{p}(x)_i} \right)$$

IPW with Normalized Weights:

$$\widehat{ATE} = \left[\sum_{n=1}^{N} \frac{Y_i * D_i}{\hat{p}(x)_i} \right] / \left[\sum_{n=1}^{N} \frac{D_i}{\hat{p}(x)_i} \right] - \left[\sum_{n=1}^{N} \frac{Y_i * (1 - D_i)}{(1 - \hat{p}(x)_i)} \right] / \left[\sum_{n=1}^{N} \frac{(1 - D_i)}{(1 - \hat{p}(x)_i)} \right]$$

2.1.1 Calculate $E[Y^1]$ and $E[Y^0]$

First, we need to calculate normalization weights

```
gen d1=treat/pscore
gen d0=(1-treat)/(1-pscore)
```

Sum the inverse pscore

```
egen s1=sum(d1)
egen s0=sum(d0)
gen total = _N
egen total_T = sum(treat)
egen total_C = sum(1-treat)
```

First, we will manually calculate $E[Y^1]$ with non-normalized weights using all the data. Calculate $E[Y^1] = \frac{1}{n} * \sum_{n=1}^{N} \frac{Y_i * D_i}{\hat{p}(x)_i}$

```
gen y1=treat*re78/pscore
egen y1_2 = sum(y1)
gen y1_3 = y1_2/total
```

Next, we will manually calculate $E[Y^0]$ with non-normalized weights using all the data. Calculate $E[Y^0] = \frac{1}{n} * \sum_{n=1}^{N} \frac{Y*(1-D_i)}{(1-\hat{p}(x)_i)}$

```
gen y0=(1-treat)*re78/(1-pscore)
egen y0_2 = sum(y0)
gen y0_3 = y0_2/total
```

Then calculate the ATE by taking the difference in $E[Y^1] - E[Y^0]$.

```
*Way 1
gen ht=y1-y0
*Way 2
gen ht2=y1_3-y0_3
*Same results
sum ht ht2
```

```
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
          ht |     16,177   -11876.79    127578.8  -99232.64   1.19e+07
         ht2 |     16,177   -11876.79           0  -11876.79  -11876.79
```

Now we will calculate the ATE manually using normalized weights

```
*Way 1
replace y1=(treat*re78/pscore)/(s1/total)
replace y0=((1-treat)*re78/(1-pscore))/(s0/total)
*Way 2
replace y1_3=y1_2/s1
replace y0_3=y0_2/s0
*Get the estimated ATE
gen norm=y1-y0
gen norm2=y1_3-y0_3
*Same Results for ht:ht2 and norm:norm2
sum ht ht2 norm norm2
```

```
(140 real changes made)

(13,820 real changes made)

(16,177 real changes made)

(16,177 real changes made)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
          ht |     16,177   -11876.79    127578.8  -99232.64   1.19e+07
         ht2 |     16,177   -11876.79           0  -11876.79  -11876.79
        norm |     16,177    -7238.14    332633.6   -99127.2   3.11e+07
       norm2 |     16,177    -7238.14           0   -7238.14   -7238.14
```

ATE under non-normalized weights is -$11,876. ATE under normalized weights is -$7,238.

Why are they so much different than the experimental research design estimates? Given that the mean $1-p(x)$ is close to 1, a lot of weight is given to Y^0. What we need to do is trim and recalculate.

2.1.2 Trim the data

We will drop our calculated variables and then trim our p-scores.

```
drop d1 d0 s1 s0 y1 y0 ht norm
drop if pscore <= 0.1
drop if pscore >= 0.9
```

We will recalculate the non-normalized weights using trimmed data.

```
gen d1=treat/pscore
gen d0=(1-treat)/(1-pscore)
egen s1=sum(d1)
egen s0=sum(d0)

gen y1=treat*re78/pscore
gen y0=(1-treat)*re78/(1-pscore)
gen ht=y1-y0
```

We will recalculate the normalized weights using trimmed data.

```
replace y1=(treat*re78/pscore)/(s1/_N)
replace y0=((1-treat)*re78/(1-pscore))/(s0/_N)
gen norm=y1-y0
sum ht norm
```

```
(0 real changes made)

(0 real changes made)

(0 real changes made)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
          ht |        361    2006.365    19656.33  -73545.45   131755.3
        norm |        361     1806.73    18967.72  -72480.23   126251.9
```

ATE under non-normalized weights is $2,006, while the estimated ATE under normalized weights is $1,807.

These are much closer to our experimental research design estimates.

2.2 Calculate ATE and ATT using *teffects*

Luckily, we have a built-in Stata command with *teffects ipw*, so we don't need to manually calculate the ATE everytime. (However, it is a good exercise to see how Stata calculate the estimate). The *teffects* command will also calculate our standard errors, as well.

We will reload our data, trim the data, and use the *teffects ipw* command. We will also trim our data, since we know that there is a lack of overlap at the tail ends of the estimated p-scores.

```
* Reload Data
use https://github.com/scunning1975/mixtape/raw/master/nsw_mixtape.dta, clear
drop if treat==0

*Now merge in the CPS controls from footnote 2 of Table 2 (Dehejia and Wahba 2002)
append using https://github.com/scunning1975/mixtape/raw/master/cps_mixtape.dta
gen agesq=age*age
gen agecube=age*age*age
gen edusq=educ*edu
gen u74 = 0 if re74!=.
replace u74 = 1 if re74==0
gen u75 = 0 if re75!=.
replace u75 = 1 if re75==0
gen interaction1 = educ*re74
gen re74sq=re74^2
gen re75sq=re75^2
gen interaction2 = u74*hisp

* Now estimate the propensity score
logit treat age agesq agecube educ edusq marr nodegree black hisp re74 re75 u74 u75 in
predict pscore

* Checking mean propensity scores for treatment and control groups
sum pscore if treat==1, detail
sum pscore if treat==0, detail

* Trimming the propensity score
drop if pscore <= 0.1
drop if pscore >= 0.9
```

With the *teffects* command, we need to specify *ipw* for inverse probability weights and we need to use the option *logit* to tell Stata to calculate the p-scores using a logistic regresion. In addition, we will use the option *osample* to look for any overlap between treatment and comparison. Finally, we will use the option *atet* to specify that we want to estimate the ATT.

```
teffects ipw (re78) (treat age agesq agecube educ edusq marr nodegree ///
black hisp re74 re75 u74 u75 interaction1, logit), osample(overlap) atet
capture drop overlap
```

```
Iteration 0:   EE criterion =  1.392e-21
Iteration 1:   EE criterion =  2.898e-23

Treatment-effects estimation                    Number of obs     =        361
```

```
Estimator      : inverse-probability weights
Outcome model  : weighted mean
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
        re78 |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ATET         |
       treat |
   (1 vs 0)  |   2646.866   892.2393     2.97   0.003     898.1089    4395.623
-------------+----------------------------------------------------------------
POmean       |
       treat |
          0  |   3791.931   514.4727     7.37   0.000     2783.582    4800.279
------------------------------------------------------------------------------
```

The ATT is estimated at $2,647

Before we calculate the ATE, we need to rescale the dependent variable for concavity of logit. We will do so by dividing re78 by 1000 dollars

The ATE is estimated at $1,611

```
gen re78_scaled = re78/1000
teffects ipw (re78_scaled) (treat age agesq agecube educ edusq marr nodegree ///
black hisp re74 re75 u74 u75 interaction1, logit), osample(overlap) ate
```

```
command quilety is unrecognized
r(199);

r(199);
```

Chapter 3

Doubly Robust Estimator

Next we will implement our doubly robust estimator. We are combining our regression adjusted outcome model and our IPW using p-scores. We have two chances to get the correct specification with the doubly robust estimator.

$$ATE_{DR} = [E[Y^1] - E[Y^0]] + E\left[\frac{(Y - E[Y^1]) * D}{p(x)} - \frac{(Y - E[Y^0])(1-D)}{(1-p(x))}\right]$$

The left-hand side is our OLS-based model and the right-hand side is our Inverse Probability Weighted Model. If we specified the OLS model correctly, our second term equals zero.

First, we will reload our data and do the trimming.

```
* Reload Data
use https://github.com/scunning1975/mixtape/raw/master/nsw_mixtape.dta, clear
drop if treat==0

*Now merge in the CPS controls from footnote 2 of Table 2 (Dehejia and Wahba 2002)
append using https://github.com/scunning1975/mixtape/raw/master/cps_mixtape.dta
gen agesq=age*age
gen agecube=age*age*age
gen edusq=educ*edu
gen u74 = 0 if re74!=.
replace u74 = 1 if re74==0
gen u75 = 0 if re75!=.
replace u75 = 1 if re75==0
gen interaction1 = educ*re74
gen re74sq=re74^2
gen re75sq=re75^2
```

```
gen interaction2 = u74*hisp

* Now estimate the propensity score
logit treat age agesq agecube educ edusq marr nodegree black hisp re74 re75 u74 u75 in
predict pscore

* Checking mean propensity scores for treatment and control groups
sum pscore if treat==1, detail
sum pscore if treat==0, detail

* Trimming the propensity score
drop if pscore <= 0.1
drop if pscore >= 0.9
```

3.1 Calculate the ATE and ATT using Doubly Robust Estimator

Our Doubly Robust Estimator using teffects includes *teffects ipwra (OUTCOME MODEL) (TREATMENT MODEL)* In the first set of paratheses, we will use the variables to estimate the regression adjustment model or outcome model. In the second set of paratheses we will estimate the treatment model or the model for to estimate our p-scores.

Next, we will calculate the ATT. Note: We have to rescale for the concavity of the logit.

```
gen re78_scaled = re78/1000

teffects ipwra (re78_scaled age agesq agecube educ edusq ///
marr nodegree black hisp re74 re75 u74 u75 interaction1) (treat age agesq agecube educ
black hisp u74 u75 interaction1, logit), atet
```

```
Iteration 0:   EE criterion =  2.996e-21
Iteration 1:   EE criterion =  6.603e-29

Treatment-effects estimation                    Number of obs     =        361
Estimator      : IPW regression adjustment
Outcome model  : linear
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
 re78_scaled |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
```

```
ATET         |
       treat |
   (1 vs 0)  |   2.565509   .8702429     2.95   0.003     .8598642    4.271154
-------------+----------------------------------------------------------------
POmean       |
       treat |
          0  |   3.873287   .4696332     8.25   0.000     2.952823    4.793752
------------------------------------------------------------------------------
```

We find that our estimate ATT is \$2,566.

Next we will estimate the ATE, and we use the option *ate* to specify that we want to estimate the ATE.

```
teffects ipwra (re78_scaled age agesq agecube educ edusq ///
marr nodegree black hisp re74 re75 u74 u75 interaction1) (treat age agesq agecube educ edusq marr
black hisp u74 u75 interaction1, logit), ate
```

```
Iteration 0:   EE criterion =  2.996e-21
Iteration 1:   EE criterion =  1.056e-28

Treatment-effects estimation                    Number of obs     =        361
Estimator      : IPW regression adjustment
Outcome model  : linear
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
 re78_scaled |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ATE          |
       treat |
   (1 vs 0)  |   1.608922   .6988993     2.30   0.021     .2391043    2.978739
-------------+----------------------------------------------------------------
POmean       |
       treat |
          0  |   4.297229   .3749753    11.46   0.000     3.562291    5.032167
------------------------------------------------------------------------------
```

We find that our estimate ATE is \$1,609.

Cunningham, S. (2021). Causal Inference: The Mixtape. Yale University Press.

Huber, M. (2023). Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R. The MIT Press.

Huntington-Klein, N. (2022). The Effect: An Introduction to Research Design and Causality. CRC Press.